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Subject: OSSP

Scheduling Algorithms

1.FCFS Scheduling algorithm

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSLAB14  Simple-Commands-Week1  foldername  hello.c  monis  sample1.c
OSLAB  OSSP-Practical  example.txt  hello  moni  sample1  sample1.c.save
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ mkdir Scheduling-Algorithms
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  file1.txt  hello.c  hello:Zone.Identifier  hello_shortcut  skill
Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd Scheduling-Algorithms
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ nano fcfs.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
GNU nano 8.7.1 fcfs.c *
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-T To
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/_ Go To Line M-E Redo M-G Copy ^B Where

Write to File: fcfs.c

^G Help

^C Cancel

M-D DOS Format

M-M Mac Format

M-A Append

M-P Prepend

Output:

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ gcc fcfs.c -o fcfs
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ ./fcfs
Enter number of processes: 3
Enter burst time:
P1: 5
P2: 3
P3: 8

Process BT      WT      TAT
P1      5        0        5
P2      3        5        8
P3      8        8       16
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$
```

2.SJF_NONpreemptive

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
GNU nano 8.7.1 sjf_nonpreemptive.c *
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    /* Sort according to burst time */
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }
}
```

Help Write Out Where Is Cut Execute Location M-U Undo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
GNU nano 8.7.1 sjf_nonpreemptive.c *
int bt[20], wt[20], tat[20];
int temp;

printf("Enter number of processes: ");
scanf("%d", &n);

printf("Enter burst time:\n");

for (i = 0; i < n; i++) {
    printf("P%d: ", i + 1);
    scanf("%d", &bt[i]);
}

/* Sort according to burst time */
for (i = 0; i < n - 1; i++) {
    for (j = i + 1; j < n; j++) {
        if (bt[i] > bt[j]) {
            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
        }
    }
}

wt[0] = 0;

for (i = 1; i < n; i++) {
    wt[i] = wt[i - 1] + bt[i - 1];
}

for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

return 0;
}

Write to File: sjf_nonpreemptive.c
^G Help          M-D DOS Format   M-A Append       M-B Backup File  ^T Browse
^C Cancel        M-M Mac Format   M-P Prepend      ^O Discard buffer
```

Output:

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ nano sjf_nonpreemptive.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ gcc sjf_nonpreemptive.c -o sjf_nonpreemptive
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ ./sjf_nonpreemptive
Enter number of processes: 4
Enter burst time:
P1: 6
P2: 8
P3: 7
P4: 3

Process BT      WT      TAT
P1      3        0        3
P2      6        3        9
P3      7        9       16
P4      8       16       24
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$
```

3.Priority

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
GNU nano 8.7.1 priority.c *
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    /* Sort according to priority */
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    /* Waiting time */
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    /* Turnaround time */
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nWaiting time: ");
    for (i = 0; i < n; i++) {
        printf("P%d: %d\n", i + 1, wt[i]);
    }

    printf("\nTurnaround time: ");
    for (i = 0; i < n; i++) {
        printf("P%d: %d\n", i + 1, tat[i]);
    }
}
```

Help	Write Out	Where Is	Cut	Execute	Location	M-U Undo
Exit	Read File	Replace	Paste	Justify	Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
GNU nano 8.7.1 priority.c *
    scanf("%d", &bt[i]);

    printf("P%d Priority: ", i + 1);
    scanf("%d", &priority[i]);
}

/* Sort according to priority */
for (i = 0; i < n - 1; i++) {
    for (j = i + 1; j < n; j++) {

        if (priority[i] > priority[j]) {

            temp = priority[i];
            priority[i] = priority[j];
            priority[j] = temp;

            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
        }
    }
}

/* Waiting time */
wt[0] = 0;

for (i = 1; i < n; i++) {
    wt[i] = wt[i - 1] + bt[i - 1];
}

/* Turnaround time */
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t\t%d\t%d\n",
        i + 1, bt[i], priority[i], wt[i], tat[i]);
}

return 0;
}

Write to File: priority.c
^G Help          M-D DOS Format  M-A Append      M-B Backup File ^T Browse
^C Cancel        M-M Mac Format  M-P Prepend     ^O Discard buffer
```

Output:

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ nano priority.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ gcc priority.c -o priority
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ ./priority
Enter number of processes: 4
Enter burst time and priority:
P1 Burst Time: 10
P1 Priority: 3
P2 Burst Time: 1
P2 Priority: 1
P3 Burst Time: 2
P3 Priority: 4
P4 Burst Time: 1
P4 Priority: 2

Process BT    Priority    WT    TAT
P1      1      1          0     1
P2      1      2          1     2
P3     10      3          2    12
P4      2      4         12    14
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$
```


4.Round Robin

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
GNU nano 8.7.1 round_robin.c *
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    /* Round Robin */
    while (1) {
        done = 1;

        for (i = 0; i < n; i++) {
            if (rem_bt[i] > 0) {
                done = 0;

                if (rem_bt[i] > quantum) {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else {
                    time = time + rem_bt[i];

                    wt[i] = time - bt[i];
                }
            }
        }
    }
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
GNU nano 8.7.1 round_robin.c *

/* Round Robin */
while (1) {

    done = 1;

    for (i = 0; i < n; i++) {

        if (rem_bt[i] > 0) {

            done = 0;

            if (rem_bt[i] > quantum) {
                time = time + quantum;
                rem_bt[i] = rem_bt[i] - quantum;
            }
            else {
                time = time + rem_bt[i];

                wt[i] = time - bt[i];

                rem_bt[i] = 0;
            }
        }
    }

    if (done == 1)
        break;
}

/* Turnaround time */
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

return 0;
}

Write to File: round_robin.c
^G Help      M-D DOS Format  M-A Append      M-B Backup File  ^T Browse
^C Cancel    M-M Mac Format  M-E Append      ^O Discard buffer
```

Output:

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/Scheduling-Algorithms
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ nano round_robin.c
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ gcc round_robin.c -o round_robin
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$ ./round_robin
Enter number of processes: 4
Enter burst time:
P1: 5
P2: 4
P3: 2
P4: 7
Enter time quantum: 2

Process BT      WT      TAT
P1      5       10     15
P2      4        8     12
P3      2        4      6
P4      7       11     18
monis@DESKTOP-C3GM4T6:~/OSLAB/Scheduling-Algorithms$
```